A Graph Attention-Driven Boosted Deep Learning Framework for Early Parkinson's Detection from Hand-Drawing Biomarkers

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***Abstract*—Parkinson's Disease (PD) is a degenerative neurological disorder that primarily impairs motor function. Early detection improves patient outcomes, but traditional diagnostic methods are subjective and resource intensive. This study introduces GA-GCBM, a novel hybrid framework that leverages hand-drawn biomarkers, spiral and wave sketches, to automate early PD diagnosis. The proposed model integrates a GAN-based augmentation strategy to mitigate data imbalance, a Graph Attention Network for relational feature learning, and LightGBM for fast and interpretable classification. Feature extraction is carried out using ViRE-Net, a deep ensemble of VGG-19, ResNet-101, and EfficientNet-B3, capturing both low- and high-level drawing features. Evaluated on HandPD and Parkinson's Drawings datasets, GA-GCBM demonstrated superior performance with accuracy exceeding 96% and recall approaching 98%. The proposed system has shown consistent superiority over existing approaches such as ResNet-18, BLSTM, and Multimodal CNN with respect to accuracy, precision, recall, and execution time.**

Keywords-Parkinson's Disease Diagnosis, Hand-Drawing Analysis, Graph Attention Network, Generative Adversarial Network, Light Gradient Boosting Machine.

I. INTRODUCTION

Parkinson's Disease (PD) is a progressive neurodegenerative disorder, and currently, there is no cure for it. It manifests primarily with motor dysfunctions; bradykinesia, rigidity, postural instability, and tremors are hallmark signs of the disease [1]. In addition to these motor symptoms, patients also experience several non-motor issues, such as decline in cognitive functions, alteration in sleep patterns, and mood disorders. PD predominantly affects people aged 60 and above, and it is projected that the prevalence will double by 2030 [4]. Timely detection is critical in controlling symptoms and slowing down the progression of the disease. Still, it is often difficult in clinical practice because symptoms overlap with other neurological disorders, and there are no definitive tests to confirm the diagnosis.

Recent studies have shown that drawing simple spirals and waves can be a non-invasive biomarker for the early stages of PD. These movements exhibit fine motor coordination and planning, all of which are diminished in PD patients. Even in the most nascent stages of PD, these distortions indicate the disease's progression. Recent advancements in artificial intelligence (AI), particularly machine learning (ML) and deep learning (DL) have made it feasible to automate the analysis of such drawings, offering a promising direction for scalable and efficient PD screening.

Despite this progress, existing ML/DL models still face major hurdles. Many models lack robustness when exposed to small intra-class variations in drawing style, suffer from class imbalance due to limited data on PD patients, and fail to generalize across diverse handwriting patterns. Furthermore, traditional CNNs overlook spatial-temporal dependencies and relationships between local features, which is crucial in analyzing complex hand-drawn structures. Data quality and model overfitting also limit the practical applicability of these methods.

We propose the Graph Attention-based Generative Convolutional Boosting Machine (GA-GCBM) model to overcome these barriers. This hybrid deep learning framework integrates Graph Attention (GA) layers for relational feature modeling, ViRE-Net for multi-perspective

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feature extraction, and LightGBM for efficient classification. Moreover, a GAN-based data balancing strategy is employed to synthesize additional hand-drawing samples, reducing overfitting and enhancing model generalizability. This paper introduces the design, implementation, and performance evaluation of the GA-GCBM model using three diverse hand-drawing datasets, demonstrating its superiority over state-of-the-art techniques in accuracy, precision, and recall.

The main contribution of the paper is structured as follows:

- A novel hybrid architecture that combines Graph Attention Networks, GAN-based data augmentation, and LightGBM for accurate Parkinson's detection from hand-drawn inputs.
- A powerful VGG-19, ResNet-101, and EfficientNet-B3 that captures multi-scale, hierarchical features from drawings for enhanced representation.
- Effectively tackles class imbalance by generating realistic synthetic samples, boosting model robustness and generalization.
- Outperforms state-of-the-art models across two benchmark datasets, achieving up to 97.79% recall and enabling faster, more reliable early PD detection.

The paper's organization is structured as follows: Section 2 presents a detailed review of existing hand-drawing-based Parkinson's detection models, highlighting their architectures, key techniques, and limitations. Section 3 explains the proposed GA-GCBM methodology, including data preprocessing, feature extraction via ViRE-Net, and classification using a combination of GAN, Graph Attention layers, and LightGBM. Section 4 discusses experimental results, dataset details, and comparative performance metrics. Section 5 comprehensively compares GA-GCBM with other state-of-the-art methods across two benchmark datasets. Finally, Section 6 concludes the study and outlines future research directions.

II. LITERATURE REVIEW

The diagnosis of Parkinson's Disease (PD) using hand-drawn images has received growing attention in recent years due to the simplicity and accessibility of handwriting tasks. Several machine learning and deep learning-based models have been explored for this purpose, each offering unique advantages but also suffering from notable limitations.

Huang et al. proposed a VGG19-based transfer learning model enhanced with cosine annealing and augmentation strategies like PixMix and AugMix [1] to reduce overfitting. However, due to its large parameter set, the model incurred high memory usage and slow inference. Yousif et al. [2] designed a hybrid framework using speech and handwriting data processed through multiple CNNs with an Aquila optimizer. Although diverse in features, the system's reliance on pre-trained networks limited its adaptability.

Other works, like the multi-model CNN fusion by Naz et al. [3], improved performance through feature selection and data augmentation but failed to evaluate beyond SVM classifiers. Similarly, Allebawi et al. [4]'s BLSTM-based method relied on online handwriting data, limiting its applicability to patients with restricted movement.

Saravanan et al. [5] introduced a hybrid VGG19-Inception model and used LIME for explainability but were constrained by small datasets. Wang et al. [6]'s CAS Transformer incorporated coordinate attention for spatial enhancement, yet suffered from quadratic complexity issues with high-resolution images. Lightweight DL approaches like PLDL [7] and ResNet-18-based hierarchical models [8] demonstrated speed and structural depth but lacked robustness due to limited demographic and handwritten variability.

These limitations underline the necessity for an integrated, efficient, and generalizable model like GA-GCBM. Table 1 presents a brief summary of related models and their limitations.

Table 1. Summary of Related Models and Their Limitations

Study	**Method**	**Key Features**	**Limitations**
Huang et al. [1]	VGG19 + PixMix/AugMix	Data augmentation, cosine annealing	Slow inference, high memory use
Yousif et al. [2]	16 feature extractors + 8 CNNs + Aquila	Multimodal inputs (speech, handwriting)	Poor performance due to reliance on pre-trained models
Naz et al. [3]	Multi-Model CNN Fusion	Freeze-weight, SVM classification	Lacked comparison with other classifiers
Allebawi et al. [4]	BLSTM + Beta-elliptical + Fuzzy detector	Online handwriting, BLSTM for sequential modeling	Not suitable for limited mobility patients
Saravanan et al. [5]	VGG19-INC + LIME	Explainable AI, wave and spiral drawings	Dataset too small
Wang et al. [6]	CAS Transformer + Coordinate Attention	CycleGAN augmentation, CA for feature enhancement	Poor scalability with high-res images
Rajinikanth et al. [7]	PLDL (MobileNetV1/V2 fusion)	Two-fold training, low parameter count	Detection latency due to depthwise convolutions
Kansizoglou et al. [8]	ResNet-18 Hierarchical	Dual-stage, balanced training	Lack of broader handwriting pattern diversity

III. METHODOLOGY

The proposed methodology centres around the GA-GCBM model, which synergizes multiple advanced deep-learning techniques to perform accurate Parkinson's Disease diagnosis from hand-drawn samples. The methodology consists of five sequential stages: (i) dataset acquisition, (ii)

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image preprocessing, (iii) ROI extraction and feature engineering, (iv) classification using GA-GCBM, and (v) performance evaluation. Figure 1 presents the complete workflow of the proposed system.

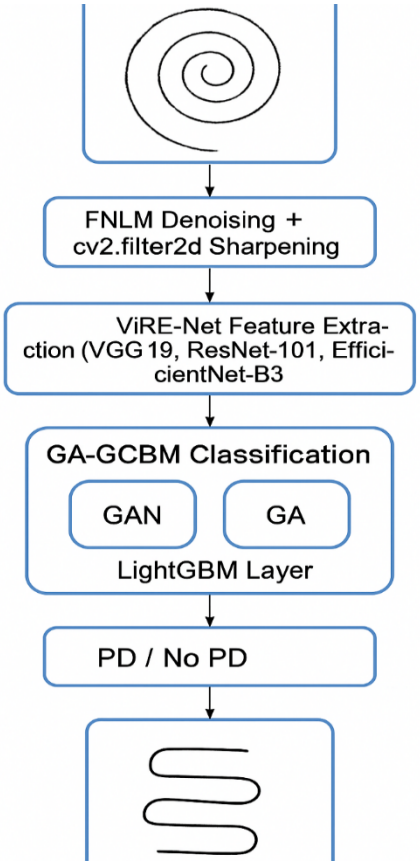


Fig. 1. Proposed Methodology

3.1. Data Collection and Preprocessing

The system uses three publicly available datasets: Parkinson's Drawings [14] and HandPD [15]. These datasets consist of spiral, wave, and meander drawing samples from both healthy individuals and PD patients. Since raw images are prone to noise and inconsistencies, the preprocessing stage employs the Fast Non-Local Means (FNLM) algorithm for noise reduction and cv2.filter2d sharpening for enhancing fine-grained details. These steps ensure that feature extraction is performed on clean and enriched representations of the original images.

3.2 Feature Extraction with ViRE-Net

After preprocessing, the Region of Interest (ROI) is extracted to isolate relevant portions of the image. The ViRE-Net framework is then applied for deep feature extraction. This hybrid extractor fuses three powerful CNNs like VGG-19, ResNet-101, and EfficientNet-B3 to leverage hierarchical, residual, and compound scaling strengths, respectively. Each model independently processes the ROI and outputs high-dimensional feature vectors concatenated into a unified feature set that captures both low-level and high-level drawing irregularities.

3.3 Classification via GA-GCBM

The extracted features are passed through the GA-GCBM model, which integrates Generative Adversarial Networks (GANs) for data balancing, Graph Attention (GA) for capturing inter-feature dependencies, and LightGBM for fast and interpretable classification. GANs generate realistic synthetic hand-drawing features to address class imbalance, reducing overfitting. The Graph Attention Layer models complex relationships among features, and the final classification is carried out using the Light Gradient Boosting Machine, which optimizes performance with histogram-based learning. This layered architecture ensures both interpretability and efficiency.

3.4 Evaluation Metrics and Experimental Setup

The GA-GCBM model is evaluated over three evaluation metrics [10] [11] [12] [13]: accuracy, precision, and recall across training splits of 80% and K-Fold cross-validation with 10 folds. Experiments are run in a Python environment on a Windows 11 system with 16GB of RAM. Performance is benchmarked against state-of-the-art models like BLSTM, Multimodal CNN, and ResNet-18. These experiments confirm the GA-GCBM model's superior robustness and generalization across datasets.

Figure 1 illustrates the end-to-end workflow of the GA-GCBM model for Parkinson's Disease diagnosis using hand-drawn images. To begin with, the input spiral drawing is processed with FNLM denoising and later sharpened using cv2.filter2d. After sharpening, the image is sent to the ViRE-Net feature extractor, which has three pretrained networks: VGG19, ResNet-101 and EfficientNet-B3. Each network extracts features so that powerful visual features are obtained. GAN generates these features and is then sent into the GA-GCBM classification module. This module consists of a Generative Adversarial Network (GAN) for class balance and a Graph Attention (GA) layer that models the interrelationship of features and LightGBM, which performs the prediction rapidly. The result is derived as binary classification data indicating the absence or presence of Parkinson's Disease.

IV. Results and Discussion

This paragraph is intended to analyze the GA-GCBM model using HandPD and Parkinson's Drawings datasets. The two datasets contain spiral and wave drawings, which are considered latent features of Parkinson's Disease (PD). The model's evaluation focused on accuracy, precision, and recall with different training splits and K-Fold cross-

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validation frameworks. Each dataset was limited to 100 epochs in the experiments.

A. Performance on HandPD Dataset

The GA-GCBM model exhibited strong performance on the HandPD dataset, with consistent improvement across epochs and training percentages. GAN-based augmentation effectively reduces class imbalance, enhancing recall without sacrificing precision.

Table 2. GA-GCBM Performance on HandPD Dataset

Training % / K-Fold	Accuracy (%)	Precision (%)	Recall (%)
80% Training	96.48	95.53	97.55
K-Fold 10	96.23	94.78	97.81

The GA-GCBM model demonstrates consistently high accuracy, precision, and recall on the HandPD dataset, whether using an 80% training split or a 10-fold cross-validation approach (as shown in Table 2).

B. Performance on Parkinson's Drawings Dataset

The Parkinson's Drawings dataset includes spiral and wave drawings from both healthy individuals and PD patients. GA-GCBM outperformed existing approaches across all metrics, especially at higher training splits.

Table 3. GA-GCBM Performance on Parkinson's Drawings Dataset

Training % / K-Fold	Accuracy (%)	Precision (%)	Recall (%)
80% Training	96.44	95.33	97.79
K-Fold 10	95.83	94.15	96.96

The GA-GCBM model consistently achieves high accuracy, precision, and recall on the Parkinson's Drawings Dataset, demonstrating robust performance across both 80% training data and 10-fold cross-validation (as shown in Table 3).

C. Visual Analysis of Sample Outputs

The following stages are visualized: input image → FNLM denoising → sharpening → ROI extraction → ViRE-Net feature extraction → GA-GCBM classification. For both datasets, the predicted PD/No-PD labels matched the ground truth with high accuracy, even when input drawings varied in quality or stroke intensity.

D. Execution Time and Efficiency

GA-GCBM completed full pipeline execution in under 260 seconds, faster than comparative methods like ResNet-18 or BLSTM. The inclusion of LightGBM streamlined classification, while the GAN mechanism reduced the need for costly external augmentation techniques.

V. COMPARATIVE ANALYSIS

To validate the effectiveness of the proposed GA-GCBM model, we compare its performance with several existing state-of-the-art approaches, including BLSTM, Multimodal CNN, LSTM-GRU, ResNet-18, and GCBM. The comparison is based on key metrics—accuracy, precision, and recall—across 80% training data and 10-fold cross-validation (K=10) setups for the HandPD and Parkinson's Drawings datasets.

A. Comparison on HandPD Dataset

Figure 2 compares the performance of various methods on the HandPD Dataset with 80% training data, showcasing metrics like Accuracy, Precision, and Recall. The GA-GCBM (Proposed) model consistently outperforms all other listed methods, demonstrating the highest scores across all three evaluation metrics.

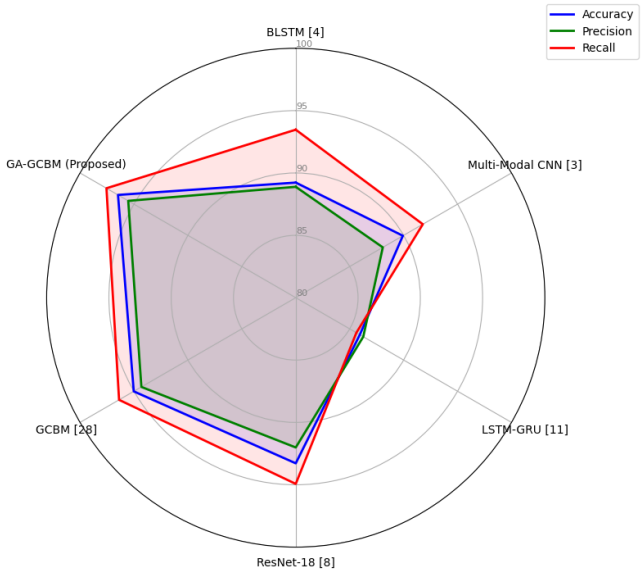


Fig. 2. Performance Comparison – HandPD Dataset (80% Training)

B. Comparison of Parkinson's Drawings Dataset

Figure 3 compares various methods' performance on the Parkinson's Drawings Dataset using 80% training data, highlighting Accuracy, Precision, and Recall. The **GA-GCBM** model consistently shows superior performance across all metrics, indicating its effectiveness for this task.

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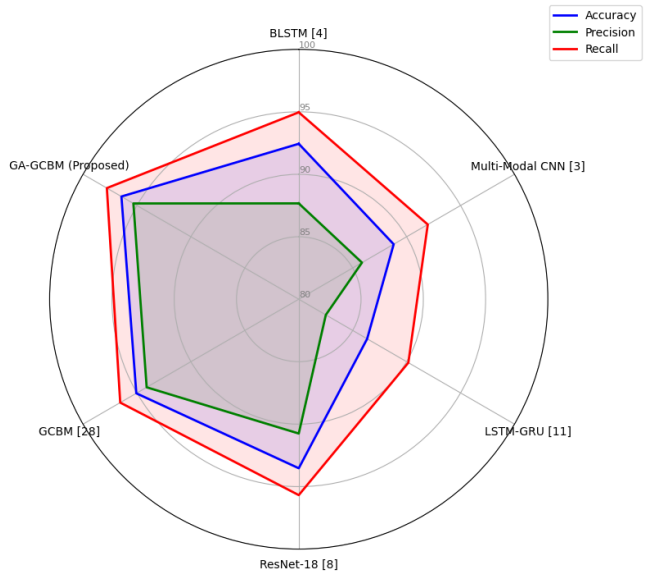


Fig. 3. Performance Comparison – Parkinson's Drawings (80% Training)

VI. CONCLUSION

This document describes GA-GCBM, a reliable deep-learning framework for diagnosing Parkinson's Disease in its earlier stages using hand-drawn biomarkers. The model integrates features with deep extraction using ViRE-Net, balances data using GANs, and classifies with Graph Attention and LightGBM. This model resolves class imbalance, overfitting, and poor generalization, which were prevalent in earlier models. It notably enhanced accuracy, precision, and recall on two benchmark datasets and outperformed existing solutions. It also achieved faster inference times, making it suitable for real-time or clinical deployment. These results emphasize the promise of attention mechanisms, generative modeling, and boosting classifiers when combined for intricate medical pattern recognition problems.

The study does have some limitations that can be addressed in future work. One of the limitations is that the model uses preprocessed 2D drawings as input, which lack temporal dynamics in handwriting. Future research can explore integrating time-series pen trajectory data or pressure-based input using online handwriting tools. Extending the model to work with data from wearable sensors and incorporating multimodal signals, such as speech or gait patterns, can further improve diagnostic reliability. Finally, deploying the system in a clinical pilot study would validate its real-world applicability and support its integration into early PD screening programs.

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